BUILDING AI APPLICATIONS · ASSIGNMENT 3

HomeStretch: Retrieval, Memory, and Data Handling

Building AI Applications, Assignment 3 · Group 5: Venkata Sriya Kamarsu, Anmol Shetty, Darshan Patil · November 2026
Design detail, scripts, test sets, and every run: **github.com/anmolshetty02/homestretch**, `docs/rag-memory/`. Underlined results link to the exact file.

1 What changes, and why

Maria cares for her mother Rosa after a heart-failure discharge. Prompt A reads labels into JSON; Prompt B is the chat agent (v4 = A2's final, v8 = this design). Each choice beat alternatives under a rule set before testing.

A2 gap	Fix	Evidence
Unsupported safety language	Facts only from the plan or a cited label `[ref-n]`; 911 replies use no label text	911 replies with advice not in the plan: v4 5 of 13, v8 0 of 13
Unknown medications	RxNorm name check before any search; no match → `pharmacist_check`	Caught 4 of 4 made-up drugs (search scores alone: 0)
Timing differences	Schedule stored as fields with the exact timing words	Fields kept "4 PM" and "bedtime"; summaries lost them
Multi-turn behavior	Structured session record: open flags, questions, warning signs	Only approach keeping every fact on both 11-turn chats

2 Knowledge sources and RAG

The care plan (~1,500 tokens) and cabinet check are small and always relevant: in the prompt every turn (cache-augmented generation, CAG: no search needed). FDA labels are large and one section matters per question: retrieved from openFDA [1] [2]. RxNorm [3], NLM's standard drug-name list, names drugs; NLM retired its interaction API [3], so interactions come from labels.

Only medicine questions retrieve (never 911 cases, dose questions, or small talk): (1) name check against the patient's list, then exact RxNav; no match → `pharmacist_check` ("ask the pharmacist"); (2) filter to her active medicines and drugs named; (3) add label words ("itchy" → pruritus), hybrid search (BM25 keyword match + vectors that match meaning); (4) a reranker (a second model that re-scores the matches) orders the top 20, top 5 kept; (5) pass them in a new `{{references}}` slot with label id and date. Prompt B v8 says "the label says [ref-1]", trusts plan over label over memory, else names the pharmacy.

Experiment (free, no LLM)	Tried	Result → decision
151 messy names on 379 bottles	Exact, approximate (4 cutoffs), fuzzy list (3 cutoffs)	Approximate matched **wrong real drugs** (`1lsinopril` → fosinopril); safe only at its strictest cutoff, which found 30 of 150: rejected. Patient's own list, then exact: 315 of 379 right, **0 wrong**
8 labels; 45 answerable + 12 no-answer questions	Chunk sizes; BM25 / vector / hybrid; reranker; expansion; top 3 vs 5	Hybrid + expansion, reranked, top 5: 34 of 37 chosen-on questions; **9 of 12 on 15 new frozen questions** (old pipeline: 2 of 12, its gate blocked 9). A stopped drug in the filter caused 5 wrong-drug answers; fixed: 0
"I don't know" gate	Similarity vs reranker score; rechecked on the final pipeline	Re-checked, the reranker gate blocked 19 of 37 real questions (the earlier 1 of 29 was another setup): **removed**. The name check stops made-up drugs; v8 ignores passages that do not answer

3 Memory

Short-term. Four ways to carry an 11-turn chat (6 scored facts), confirmed on a held-out chat (written after the choice, never used to choose):

Approach	Safety facts kept (dev / held-out)	Other facts	Size
Last 3 or 5 turns	1–2 / 1 of 3	1 / 1 of 3	35–55 words
80-word summary	3 / 3 of 3	2 / 1 of 3	~70 words
Structured record	**3 / 3 of 3**	**3 / 3 of 3**	~260 words

Windows dropped warning signs; summaries blurred times (though they kept all facts on two 20-turn chats). The record fills a new `{{memory}}` slot each turn: exact ids and sources, no small talk, flags closed only on evidence; caregiver-reported changes stay questions, never plan edits.

Long-term: semantic (confirmed medicines, schedule), episodic (weight log, past flags, rechecks), procedural (reply preferences); confirmed writes only; view, edit, delete; deleted at day 60 (30-day program + 30 to decide). In two new-session tests, v8 raised every saved follow-up in all 3 runs; the baseline raised none. A resolved flag also updates the cabinet check, so it never comes back.

4 Integration and scenario

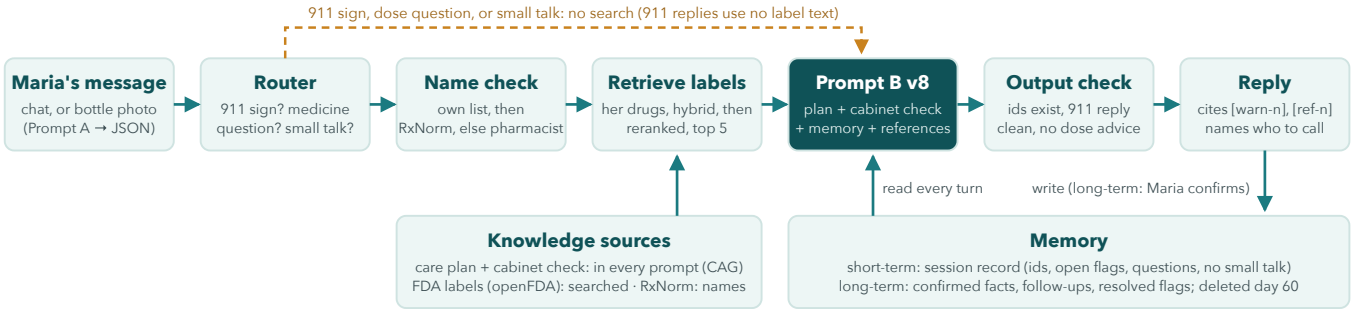


Figure 1. One message through HomeStretch. Search runs only for medicine questions; memory is read every turn and written after the reply.

Real runs [11] on synthetic Rosa; baseline = Prompt B v4, no memory or retrieval. After the chat above, Maria asks *"I'm about to call Brighton Pharmacy. What should I ask them?"* (x/3 = runs that raised it; all replies)

Earlier in the chat	Baseline (v4)	HomeStretch (v8 + memory)
Old Lopressor already bagged	calls it "still here" (wrong, 3/3)	never repeated
Unknown "Zorvexa" bottle	0/3	"What is the Zorvexa 10 mg bottle?" 3/3
New furosemide not ready	0/3	asks when, and what until then, 3/3
Doctor said lisinopril at bedtime	0/3	asked, against the plan's 8 AM, 2/3
Ankles more swollen	0/3	"Call the clinic today" (her plan's warning sign) 3/3

"Does the water pill make her lose potassium?" (retrieval works here): v4: "The care plan does not say anything about potassium. I cannot answer that." v8 (3/3): "The label says potassium supplements or diet changes may be needed [ref-3]. Her plan does not say to add anything, so ask her clinic or pharmacist." When retrieval failed, v8 used none of it.

Safety. A2 suites: v8 kept every gate (911 recall 100%, 0 critical misses, 0 under-triage, i.e. never less urgent than needed); held-out 100% (v4: 98%), stress 96% (89%), main 98%. **Blind set** (32 cases by an agent that never saw the prompts, run once): v8 = v4, 2 real errors in 80, 0 critical, but 911 recall 95%: both sent a played-down "can't talk without stopping for air" to urgent care. A fix (v10) did not change it; v8 sent all 18 fresh played-down emergencies to 911, so v8 stays.

5 Data, compliance, and risks

The class build uses synthetic data only (full analysis). A parent's medicines and symptoms are likely not PHI (health data HIPAA protects) when the family adopts the app directly [4], but just as sensitive; a hospital deployment needs BAAs (HIPAA contracts), hosting and model providers included [5]. The FTC Health Breach Notification Rule likely applies;

unauthorized sharing is a breach [6]. Washington and Nevada require opt-in consent and deletion on request; Connecticut bars selling consumer health data without consent [7–9]. DeepSeek stores data in China [10]: real data needs a self-hosted or BAA-covered model.

Access. Each person sees only what their role needs:

Who	Sees	Can do
Rosa (patient, account owner)	everything	invite or remove caregivers; export; delete all
Maria (primary caregiver, invited by Rosa)	plan, cabinet check, chat, memory	confirm, edit, or delete memory entries
Other family (invited)	plan and open items	read only
Clinic (hospital deployment only, under a BAA)	a summary Rosa chooses to share	read only
Model provider	one turn's prompt	nothing kept: a requirement for any real deployment (DeepSeek offers no BAA, so: self-hosted)

Controls: a consent screen before the first upload (what is stored, where, for how long); encryption in transit and at rest; an audit log of every view, edit, and export; every memory entry shows its source and date and can be deleted on its own; everything is deleted at day 60 unless Rosa renews; a breach means notifying users and the FTC [6].

Risk	Mitigation
Wrong, outdated, or invented drug facts	Name check; plan or cited, dated label only; stopped drugs only when named
Memory poisoning, injection, staleness	Source and date per entry; confirmed writes; documents are data, not instructions
Over-reliance	Never diagnose or dose; always name who to call

Limits: retrieval misses 3 of 12 new questions (2 ranked too low, 1 stopped by the name check); a borderline breathing sign can be under-triaged (blind set); replies run long; some fixes were picked after seeing results (labelled in the repo).

References (all checked 2026-11-01)

[1] U.S. FDA, openFDA Drug Labeling API, open.fda.gov/apis/drug/label
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[3] U.S. NLM, RxNav news and Terms of Service, lhncbc.nlm.nih.gov/RxNav
[4] U.S. HHS, Resources for Mobile Health Apps Developers, hhs.gov/hipaa/for-professionals/special-topics/health-apps
[5] U.S. HHS, Business Associates, hhs.gov/hipaa/for-professionals/privacy/guidance/business-associates
[6] U.S. FTC, HBNR changes (2024), ftc.gov/news-events/news/press-releases/2024/04/ftc-finalizes-changes-health-breach-notification-rule
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[9] Connecticut, Public Act 23-56, cga.ct.gov/2023/act/Pa/pdf/2023PA-00056-R00SB-00003-PA.PDF
[10] DeepSeek, Privacy Policy (Feb 10, 2026), cdn.deepseek.com/policies/en-US/deepseek-privacy-policy.html
[11] AI tools: Claude Code (research, scripts, drafting); DeepSeek deepseek-flash; bge-small-en-v1.5; ms-marco-MiniLM-L-6-v2